

Annual Examination 2026
Biology – Class IX | Paper: SSC-I
Syllabus: Chapter 7 – Metabolism

Total Marks: 65

Time: 2h 30m

Version: 1

General Instructions:

- Write your Roll Number in the space provided on the answer sheet. Do NOT write your name.
- Attempt ALL questions from Section A. In Sections B and C, attempt the required number choosing either the main question OR its alternative (OR).
- All parts of a question must be answered in sequence and at one place.
- Extra attempt of any question or any part thereof will NOT be considered.

SECTION A – Multiple Choice Questions (12 × 1 = 12 Marks)

#	Question	A	B	C	D	Answer
1	The sum of all chemical reactions in a living cell is called:	☐ Catabolism	☐ Anabolism	☒ Metabolism	☐ Enzymology	
2	Enzymes are biologically active globular proteins produced by:	☐ Dead cells	☒ Living cells	☐ Inorganic ions	☐ Ribosomes only	
3	The part of an enzyme where substrate binds and reaction takes place is the:	☐ Allosteric site	☐ Prosthetic group	☒ Active site	☐ Coenzyme	
4	At very high temperature, an enzyme loses function because its structure is:	☐ Activated	☒ Denatured	☐ Phosphorylated	☐ Synthesised	
5	In competitive inhibition, the inhibitor blocks enzyme activity by binding to the:	☐ Allosteric site	☐ Substrate	☒ Active site	☐ Cofactor	
6	ATP is called the "energy currency" of the cell because it:	☐ Is in the nucleus	☒ Stores and releases energy	☐ Builds proteins	☐ Carries oxygen	
7	ATP molecule consists of adenine, ribose sugar and:	☐ Two phosphate groups	☒ Three phosphate groups	☐ One phosphate group	☐ Four phosphate groups	
8	During photosynthesis, oxygen is released as a by-product through:	☐ Calvin cycle	☒ Photolysis of water	☐ Glycolysis	☐ Krebs cycle	
9	Light-independent reactions of photosynthesis (Calvin cycle) occur in the:	☐ Thylakoid membrane	☐ Cytoplasm	☒ Stroma of chloroplast	☐ Mitochondrial matrix	
10	Glycolysis produces 2 pyruvic acid from glucose with a net gain of:	☐ 36 ATP	☐ 4 ATP	☒ 2 ATP	☐ 38 ATP	
11	Anaerobic respiration in yeast (alcoholic fermentation) produces ethyl alcohol and:	☐ Water	☐ Lactic acid	☒ Carbon dioxide	☐ Oxygen	
12	The final electron acceptor in the electron transport chain (ETC) is:	☐ NADH	☐ Carbon dioxide	☒ Oxygen	☐ Water	

Q: 2(i)

A: 2(i)

Metabolism

The word metabolism means to change. The sum of all the chemical reactions going on within each cell of living organisms for the maintenance and sustainability of life is known as metabolism.

CatabolismAnabolismDefinition

* Biochemical reaction which break down complex molecules into simpler molecules called catabolic reactions or catabolism.

Reactions which build complex molecules from simpler molecules are called anabolic reactions or anabolism.

Examples

* Digestion is the catabolic process that turns larger and complex food into simpler and small molecules of glucose.

* Condensation of glucose molecules to form glycogen in muscle and liver cells is an anabolic reaction.

Q: 2(ii)(or)

A: 2(ii)(or)

✓ Enzymes are specific:-

Enzymes are usually very specific so catalyse only one type of reaction and it will not act on a different substrate.

FOR EXAMPLE:-

Amylase will only act on starch and not on proteins or fats.

Q: 2(iii)(or)

A: 2(iii)(or)

Energy of activation:-

Minimum amount of extra energy required to activate the molecule to react or start the reaction is called **energy of activation**.

Enzymes lower the activation energy so that these reactions can take place at body temperature.

① Enzymes lower the activation energy by changing the shape and charge of the substrate.

② Enzyme may provide entirely different mechanism of reaction to lower the activation energy.

— Q:2 : (iv) —

— A:2 (iv) —

Enzymes on high temperature:-

At very high temperature atoms of the enzyme molecule starts vibrating vigorously so, globular structure and active site of enzyme is damaged and it is called denatured.

Enzymes on Low temperature:-

At very low temperature enzyme remain inactive due to fall in available activation energy.

— Q:2(v) —

— A:2(v) —

Competitive Inhibition	Non-competitive inhibition
Defination	

- | | |
|--|--|
| <p>* If enzymes are inhibited due to the presence of molecules which occupy the active site due to their similarity in shape or structure to substrate is called competitive inhibition.</p> | <p>* The structure and shape of these inhibitors do not resemble the substrate molecule so they do not occupy the active site.</p> |
|--|--|

Reversibility

- | | |
|-------------------------------|---------------------------------|
| <p>* They can be reserved</p> | <p>They cannot be reserved.</p> |
|-------------------------------|---------------------------------|

Q:2(vi)

A:2(vi)

ATP molecule consists of three components:-

- 1) Adenine (double ringed nitrogen base)
- 2) Ribose (Five carbon sugar)
- 3) Three phosphate groups (PO_4) linked in a chain called triphosphate group.

Adenine nitrogen base binds to ribose sugar and become adenosine.

Adenosine = Adenine + 5 Carbon sugar $\rightarrow$ Adenosine.

First phosphate group binds with adenosine to form AMP. Second phosphate group binds with AMP to form ADP. Third phosphate group binds with ADP to form ATP.

Q:2(vii)

A:2(vii)

Equation :- $6\text{CO}_2 + 12\text{H}_2\text{O} \xrightarrow[\text{chlorophyll}]{\text{light}} \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{H}_2\text{O} + 6\text{O}_2$

Raw materials :- CO_2 (from air (stomata)) and H_2O (absorbed by roots). Chlorophyll (captures light energy).

Products :- Glucose (used for energy etc) and Oxygen (by product released through stomata).

Conditions :- Light energy and chlorophyll are essential.

Photosynthesis occurs in plants, algae and cyanobacteria.

Q:2(vii)

A:2(viii)

Defination :-

Reactions that require light energy directly; they depend on photosystems I and II to absorb light and begin ~~the~~ photosynthesis.

Role I :-

Electrons released by photolysis replace electrons lost from the excited chlorophyll of photosystems II and maintaining the reaction.

Role II :-

Oxygen atoms are released as O_2 through stomata, H^+ ions reduces $NADP^+$ to NADPH.

Q: 2 (in) ———
 A: 2 (in) ———

1) Glycolysis (cytoplasm) :-
 $\text{glucose} \rightarrow 2 \text{ pyruvic acid} + 2 \text{ ATP} + 2 \text{ NADH}$

2) Formation of ACETYL CoA (mitochondria) :-
 $2 \text{ pyruvic acid} \rightarrow 2 \text{ Acetyl CoA} + 2 \text{ CO}_2 + 2 \text{ NADH}$

3) KREBS Cycle :- $2 \text{ Acetyl CoA} \rightarrow 4 \text{ CO}_2 + 2 \text{ ATP} + 6 \text{ NADH} + 2 \text{ FADH}_2$

4) Electron transport chain :- $\text{NADH} + \text{FADH}_2 \rightarrow 32 \text{ ATP} + \text{H}_2\text{O}$

Q: 2 (in) ———
 A: 2 (in) ———

Definition:

First stage of cellular respiration, one glucose (6C) broken down into two molecules of pyruvic acid (3C each).

Location:-

Cytoplasm of the cell, outside the mitochondria.

Products:-

$2 \text{ pyruvic acid} + 2 \text{ ATP (net)} + 2 \text{ NADH}$

Q:2 (ni/or)

A:2 (ni/or)

- 1) Early Earth had no free oxygen; earliest organisms produced energy by anaerobic respiration.
- 2) Bacteria use anaerobic respiration for cheese and food yogurt making.
- 3) ~~Yes~~ Yeast fermentation used in wine making and baking industry.
- 4) Lactic acid fermentation provides emergency energy to muscles cells during vigorous exercise when oxygen supply is insufficient.

Section - C

Q:3 (i)

A:3 (i)

Enzymes :- Enzymes are biological catalysts which are biologically active globular proteins made by living cells, which speed up extremely biochemical reaction (without being used).

Enzymes have following characteristics:

1) Enzymes are globular proteins.

All enzymes are made up of amino chains of amino acids ~~fold~~ folded to form 3D protein molecule called globular proteins. Few enzymes are RNA in nature called ribozymes.

2) Enzymes are specific:-

Enzymes are usually very specific so catalyze only one type of reaction and it will not act on a different substrate. For Example :- Amylase will only act on starch and not on protein.

3) Enzymes are required in small quantity:-

Structure of enzymes may slightly change during reaction but they regain their actual shape, so they remained unchanged at the end of the reaction. So they can be used repeatedly. Thus, a very small ~~amount~~ quantity of an enzyme is capable of catalysing a huge amount of substrate.

4) Regulations of enzyme production and activity.

Synthesis of enzymes within the cells and can be increased or decreased by a cell according to requirements. Enzyme activity can be regulated by inhibitors or activators within or from out of cell.

5) Enzymes make metabolic energy:-

Enzymes work together in a specific sequences to make a metabolic pathways. Metabolic pathway is a series of connected chemical reactions where multiple intermediate molecules are produced before conversion of an initial substrate into final product.

— Q:3(ii)(or) —

— A: 3(ii)(or) —

Enzyme inhibition:-

Chemicals at the reaction site prevent enzymes from performing their role. An inhibitor binds to the enzyme in place of the substrate but does not convert to products. External factors:- poisons, cyanide, antibiotics, drugs etc.

Competitive inhibition :-

Inhibitor's structure resembles substrate $\rightarrow$ occupies ~~at~~ active site, substrate blocked that's why there's no product.

It is reversible; excessive substrate displaces inhibitor,

Example :-

Sulfonamide drugs competitively inhibit bacterial enzyme needed for folic acid synthesis and then it kills bacteria (antibiotic action)

Non-competitive inhibition :-

Inhibitors that does not resemble substrate, binds to ~~allosteric~~ ~~at~~ allosteric site (enzyme surface, not active site), alters globular shape, active site then damaged ~~that~~ then the substrate cannot bind, then reaction stops - ~~Irreversible~~ It is irreversible.

Example :-

Cyanide (rat poison) inhibits cytochrome c oxidase in ETC. Heavy metals (Pb, Hg) can also act as non-competitive inhibitors.

~~Q:3 (iii) (or)~~

A:3 (iii) (or)

It takes place in plants, algae, cyanobacteria. Chlorophyll absorbs light; CO_2 and H_2O are raw materials; glucose and O_2 are products.

Light - dependent reaction (Phase I) :-
In thylakoid membrane (grana).
Starts at Photosystem II :-

- 1) Chlorophyll absorbs light then emits electron pair and then it accepted by primary electron acceptor.
- 2) ~~Photo~~ Photolysis of water (H_2O) $\rightarrow$ $\text{O}_2 + 2\text{H}^+ + 2\text{e}^-$ (O_2 released through stomata).
- 3) Electron pair pass through ETC - energy for ATP synthesis.
- 4) Photosystem I :- electrons + 2H^+ reduce $\text{NADP}^+ \rightarrow \text{NADPH}$.
Products :- ATP, NADPH etc.

★ **Calvin cycle (Phase 2):**
 In stroma of chloroplast. Uses ATP and NADPH from phase 1. Steps:-

1) $\text{CO}_2 + \text{RuBP} (5\text{C}) \rightarrow 6\text{C intermediate} \rightarrow 2 \times 3\text{-PGA}$

2) 3-PGA reduced by NADPH + ATP $\rightarrow$ G3P [3C sugar]

3) G3P $\rightarrow$ glucose synthesis + RuBP regeneration for next cycle.
 It is described met by melvin cycle.
 Calvin.

— G:3(iv) —

Aerobic Respiration

Anaerobic respiration

Oxygen needed

It needs oxygen

It does not need oxygen

Glucose oxidation

Complete $\rightarrow \text{CO}_2 + \text{H}_2\text{O}$

Incomplete $\rightarrow$ Alcohol / Lactic acid

ATP yield

36 ATP

2 ATP

Site	
Cytoplasm + mitochondria	only cytoplasm.

Glycolysis (first stage) :-

Both pathways start with glycolysis in cytoplasm. Glucose ($6C$) $\rightarrow$ 2 pyruvic acid ($3C$) + 2 ATP + 2 NADH

Fate in Anaerobic Respiration

Pyruvic acid not fully oxidised

Yeast : Pyruvic acid $\rightarrow$ Ethyl alcohol + CO_2
(alcoholic fermentation - wine, bread)

Muscles: Pyruvic acid $\rightarrow$ lactic acid
(lactic fermentation - yogurt, exercise fatigue).

Total ATP = only 2 (from glycolysis)

Fate in aerobic respiration :-

Pyruvic acid enters mitochondria $\rightarrow$ fully oxidised in 3 further stages.

Step 2 :- Pyruvic acid + 2 CoA $\rightarrow$ 2 Acetyl CoA + 2 CO_2 + 2 NADH.

Day: _____

Date: _____

Step 3 (Krebs) :- $2 \text{ Acetyl CoA} \rightarrow 4 \text{ CO}_2 + 2 \text{ ATP} + 6 \text{ NADH} + 2 \text{ FADH}_2$

Step 4 (ETC) :- $10 \text{ NADH} + 2 \text{ FADH}_2 \rightarrow 32 \text{ ATP} + \text{H}_2\text{O}$ (O_2 - final acceptor)

Total = $2 + 2 + 32 = 36 \text{ ATP}$
per glucose molecule.